

ALERT IN JAVASCRIPT

1.What is alert?

A popup box that appears on the screen

2.Code Syntax :

```
alert("Your message here");
```

Whatever you write inside " " will show in the box

3.Example:

```
alert("Hello Mehnil");
```

A box will popup with Hello Mehnil

4. Why Use it ?

To show a message, warning ,or test if code is working

Variable for Strings

1. Strings store text data in JavaScript
2. Text must be inside " " quotes

Example: `var name = "Mehnil";`

Variables for numbers

1.What are number variables?

variables that store numeric values like age , marks, quantity In JavaScript

2.Code Syntax :

```
let age = 25;
```

Numbers are written without quotes " "

3.Why use it?

To do math calculations like +,-,*,/ with numbers

Variable names: Legal and Illegal

1.What are Variable names?

Names we give to variables to store data in JavaScript

2. Legal names - Rules:

Start with letter or underscore _.Numbers allowed but not at start

Example: studentName, user _age, _total1 are correct

3. Illegal names - Rules:

Cannot start with number no spaces, no special keywords

Example: 1stPlace, user name, while are wrong

Math Expressions: Familiar operators

1. What are operators?

Symbols use to do math in JavaScript like +, -, *, /

2. Four main operators:

Addition +, Subtraction -, Multiplication *, Division /

Example: $5 + 3 = 8$, $10 * 2 = 20$

3. How to use them?

We use them for numbers or number variables

Example: $4 + 1$, $6 / 2$

Math Expressions: Unfamiliar operators

1. What are Unfamiliar operators?

In JavaScript, some math symbols are not used in everyday math

Example: %, ++, --

2. What do These operators do?

Each operator has a special job

1 % (Modulus) Gives the remainder after division

2 ++ (Increment) Adds 1 to any number

3 -- (Decrement) Subtracts 1 from any number

3. Code Examples:

The Modulus operator returns the remainder

For Example : $7 \% 3 = 1$

The Increment operator adds 1 to a value

For Example : let $x = 4$; $x++$ makes $x = 5$

The Decrement operator Subtracts 1 from a value

For Example : let $y = 6$; $y--$ makes $y = 5$

Prompts

1. What is Prompt?

Shows a popup box to get user input

Example: `let age = prompt ("Name?");`

2. How it works ?

"User types answer clicks Ok value stored"

Syntax: `let data = prompt (" Text?");`

3.Example :

`let age = prompt ("Age?");`

if user enters 18, then `age = "18"`

Note: Always returns text, not a number

If statements in JavaScript

1. What is if statement?

JavaScript has a special statement used to make decisions in code

It runs a block of code only when a condition is true

Example: `if (marks >= 50 ) { . . }`

2. How does if statement work?

Check condition program first look at the condition `in( )`

Runs if true if the condition is true code inside `{ }` runs

Skips if false if the condition is false code inside `{ }` skipped

3. Why is if statement important?

Smart programs Makes a code take decisions like a human

Avoids error Runs code only when it is safe or needed

User experience helps create apps that respond to user actions

Comparison operators in JavaScript

1. What are comparison operators?

comparison operators are used to compare two values in JavaScript

They check if one value is greater, smaller or equal to another

Example: >, <, ==, ===, !=

2. What Result do they give?

Boolean output always return true or false

Used in conditions Mostly used in if - else statements to make decisions

Example: age >= 18 gives true if age is 18 or more

If - else if statements in JavaScript

1. Purpose of If - Else If

if else if is used when we need to check multiple condition
step by step

It allows the program to test more than

One situation in a specific order

2. How it Executes?

JavaScript checks condition from top to bottom

The first condition that returns true will run its code block,
and all remaining conditions are skipped

Testing sets of condition in JavaScript

1. Logical AND && or Logical Or ||

logical operators are used to check multiple conditions in same time

AND&& Returns true only if all conditions are true

Or || Returns true if any one condition is true

Example: `age >= 18 && <= 60`

2. Making complex decisions

Using these operators ,we can write complex decisions in a single if statement

Instead of using many if - else, we combine conditions for cleaner code

Example: `if ( marks > 50 && attendance > 75 ) { ... }`